

# Independent Software Support System Evidence and Operations Specification

Data definitions, calculations, payment evidence, conversion measurement, and operating controls

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Status: Product design baseline

Date: 4 September 2026

## Purpose

This specification defines what the support system may count, calculate, display, and report. Its purpose is to prevent persuasive copy from outrunning the evidence. Every user facing number must resolve to a stored observation, an identified external source, or a calculation whose inputs are visible.

## Evidence classes

| Class                 | Definition                                                                                  | Display rule                                                    |
|-----------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Observed product fact | A product records a completed event or bounded duration using an approved definition        | May be stated directly with its coverage period                 |
| External market fact  | An approved source reports a menu item, price, serving mass, location, and observation time | May be called current only inside the approved freshness period |
| Derived fact          | A deterministic calculation combines identified inputs                                      | Show inputs, units, rounding, and method                        |
| Estimate              | A value uses sampling or an assumption, such as grams per fry                               | Label as estimated and retain method, sample, and date          |
| Business hypothesis   | A proposed design or timing choice expected to affect conversion                            | Do not present as a fact; assign an experiment                  |
| Unknown               | Evidence is absent, stale, ambiguous, or incompatible                                       | Omit the claim or state that it is unavailable                  |

## Canonical local history

| Field              | Meaning                                                             | Source of truth                      |
|--------------------|---------------------------------------------------------------------|--------------------------------------|
| schemaVersion      | Version used to migrate the local record                            | Support core                         |
| firstObservedAt    | Earliest time covered by this record                                | Support core                         |
| lastObservedAt     | Latest persisted activity time                                      | Support core                         |
| meaningfulSessions | Count of sessions satisfying the product definition                 | Product adapter through core reducer |
| activeMilliseconds | Sum of bounded active intervals                                     | Support core                         |
| facts              | Approved product metrics with value, unit, coverage, and provenance | Product adapter                      |
| appealHistory      | Presentations, dismissals, opt out, and last state                  | Support core                         |
| edition            | Free, paid, supporter, institutional, or development                | Build configuration                  |

## Session and active time rules

A meaningful session is defined by the product. The support core enforces interval integrity but does not decide that opening an application constitutes use. Each adapter documents its start signal, qualifying condition, idle rule, stop signals, reconnect behavior, and crash checkpoint interval before release.

| Rule            | Required evidence                                                      |
|-----------------|------------------------------------------------------------------------|
| Session start   | Timestamp and product supplied reason                                  |
| Qualification   | The event or duration that changes a launch into a meaningful session  |
| Active interval | Monotonic start and stop readings plus stop reason                     |
| Idle exclusion  | Documented threshold or an explicit product pause signal               |
| Crash recovery  | Last durable checkpoint; the unknown remainder is excluded             |
| Coverage        | The earliest date from which totals are complete                       |
| Migration       | Source schema, transformation, and whether earlier use remains unknown |

## Product fact contract

| Field           | Meaning                                                                            | Source of truth                |
|-----------------|------------------------------------------------------------------------------------|--------------------------------|
| factId          | Stable machine identifier                                                          | Product adapter                |
| labelKey        | Approved localized label                                                           | Product content                |
| value           | Integer or decimal value                                                           | Recorded product state         |
| unit            | Sessions, milliseconds, items, pages, reports, campaigns, or another approved unit | Product definition             |
| coverageStart   | Earliest included observation                                                      | Local history                  |
| coverageEnd     | Latest included observation                                                        | Local history                  |
| completeness    | Complete, partial, or unknown                                                      | Adapter validation             |
| provenance      | Event stream, save state, database, or other named source                          | Product adapter                |
| displayPriority | Relative importance within that product                                            | Approved product configuration |

The renderer may sort and format facts. It may not convert an unavailable value to zero, infer causal benefit, estimate time saved, or write a narrative event that the product did not record.

## Fact selection

The page starts with active time and meaningful sessions when both are complete. It then displays a limited set of product facts. Selection is deterministic so the same history does not produce arbitrary claims.

1. Reject facts whose completeness is unknown.
2. Reject facts outside the current user or profile scope.
3. Reject facts whose definitions have changed without a valid migration.
4. Rank remaining facts by product configured display priority.
5. Use stable fact identifiers to break ties.
6. Display only the number of facts the approved layout can present without crowding.
7. State partial coverage beside the affected fact.

## United States price study

| Field   | Meaning                                  | Source of truth |
|---------|------------------------------------------|-----------------|
| studyId | Stable identifier for the released study | Study registry  |

| Field             | Meaning                                                               | Source of truth           |
|-------------------|-----------------------------------------------------------------------|---------------------------|
| targetPopulation  | The United States restaurant prices the estimate intends to represent | Approved study plan       |
| samplingFrame     | The list or method from which restaurants are selected                | Approved study plan       |
| sampleMethod      | Selection and any geographic or market stratification                 | Approved study plan       |
| observationWindow | Named date or disclosed date range                                    | Observation record        |
| channel           | Counter, pickup, delivery, or another defined ordering channel        | Observation record        |
| taxTreatment      | Whether reported prices include tax and how this is handled           | Study plan                |
| sampleSize        | Number of valid restaurant observations                               | Derived from observations |
| missingCount      | Selected observations that were unavailable or excluded               | Study record              |
| summary           | Mean, median, minimum, maximum, and any approved weighting            | Reproducible calculation  |
| sourceArchive     | Underlying restaurant, product, price, source, and timestamp records  | Evidence archive          |

The displayed average must name its statistic. An arithmetic mean is the sum of valid observed prices divided by the number of valid observations. Prices from incompatible channels, currencies, tax treatments, or product configurations are not combined. A result without its observations and method cannot ship as a current United States average.

## Fry mass evidence

| Field                 | Meaning                                                            | Source of truth                                     |
|-----------------------|--------------------------------------------------------------------|-----------------------------------------------------|
| menuItemId            | Canonical fries item and serving size                              | Approved item mapping                               |
| servingMassGrams      | Mass of the priced serving                                         | Official nutrition source or documented measurement |
| servingMassSource     | Source, market, and observation date                               | Evidence record                                     |
| sampleFryCount        | Number of fries weighed                                            | Measurement record                                  |
| sampleMassGrams       | Combined mass of sampled fries                                     | Measurement record                                  |
| estimatedFryMassGrams | Sample mass divided by sample count                                | Derived value                                       |
| measurementMarket     | Market in which the sample was obtained                            | Measurement record                                  |
| measuredAt            | Measurement date                                                   | Measurement record                                  |
| method                | Sampling, scale precision, inclusion of fragments, and repetitions | Measurement protocol                                |

## Fry calculation and rounding

Let P be the dated United States average serving price in major currency units, F be the estimated mass of one fry in grams, and S be the serving mass in grams. First calculate cost per gram as P divided by S. Then multiply cost per gram by F.

$$\text{cost per gram} = P / S \quad \text{estimated fry value} = (P / S) \times F$$

The system retains the unrounded result for explanation and minimum calculations. It rounds only when producing a currency amount for display or checkout. The displayed expansion substitutes the actual recorded inputs, states their dates and sources, and preserves their units.

| Provisional 4 September 2026 input | Value                              | Source basis                                                             |
|------------------------------------|------------------------------------|--------------------------------------------------------------------------|
| P: medium-fries national mean      | \$3.56                             | MenuPriceTracker, 13,595 reported U.S. locations, updated 27 August 2026 |
| Experiment basis                   | 10 orders; mean 112 g and 78 fries | Independent 2026 experiment across five franchises in one city           |

| Provisional 4 September 2026 input | Value                                                             | Source basis                         |
|------------------------------------|-------------------------------------------------------------------|--------------------------------------|
| F: estimated mass per fry          | 112 g / 78 = 1.435897 g                                           | Derived                              |
| Estimated fry value                | $(\$3.56 / 112 \text{ g}) \times 1.435897 \text{ g} = \$0.045641$ | Derived; equivalent to $\$3.56 / 78$ |
| User-facing comparison             | about \$0.05                                                      | Rounded to the nearest cent          |

The fry result is provisional because its mass and count come from a small independent experiment rather than an official portion specification or a measurement retained by this business. The implementation may display it only with that qualification until a replacement study is completed.

| Validation                                          | Failure behavior                                                    |
|-----------------------------------------------------|---------------------------------------------------------------------|
| P is positive and uses the snapshot currency        | Disable the fry calculation                                         |
| F is positive and has a measurement record          | Disable the fry calculation                                         |
| S is positive and matches the priced serving size   | Disable the fry calculation                                         |
| Price study is current under the publication policy | Label as historical or complete a new study                         |
| Currency supports the rounded amount                | Show the calculated value but use the provider minimum for checkout |
| Provider minimum is known                           | Do not offer an impossible checkout                                 |

## Contribution option record

| Field        | Meaning                                                    | Source of truth                          |
|--------------|------------------------------------------------------------|------------------------------------------|
| optionId     | Stable food or custom option identifier                    | Support core                             |
| displayLabel | Defined United States menu label or approved generic label | Price study mapping                      |
| amount       | Exact checkout amount                                      | Published price study or user input      |
| currency     | Checkout currency                                          | Published price study or payment adapter |
| basis        | Menu price, derived fry value, provider minimum, or custom | Support core                             |
| observedAt   | Observation date of the underlying price study             | Price study                              |
| availability | Available, unavailable, stale, or unsupported              | Price-study registry                     |
| recommended  | Whether this is the configured primary request             | Business configuration                   |

## Wallet commitment

The design treats wallet opening as a measurable funnel transition. The business hypothesis is that a person who has decided to open a payment method may accept a meal sized software contribution more readily than a trivial transaction. This document assigns no conversion lift and no fixed amount to that hypothesis. The checkout amount must still be shown and confirmed explicitly.

| Interaction                     | What it means                                         | Permitted next step                                     |
|---------------------------------|-------------------------------------------------------|---------------------------------------------------------|
| View fry calculation            | Curiosity or evaluation; no payment intent is assumed | Expand the method or select a contribution              |
| Select fry value                | Interest in the smallest reference amount             | Explain the provider minimum or allow accumulation      |
| Open payment method             | A stronger signal of intent                           | Show a final explicit amount and purpose before charge  |
| Confirm meal sized contribution | The user knowingly accepts the stated charge          | Create provider checkout                                |
| Cancel                          | The user did not complete payment                     | Return without recording revenue or increasing pressure |

The system may test whether the fry calculation leads more people to a meal sized contribution. It may not label a meal charge as one fry, preselect an undisclosed amount, or interpret a calculation expansion as consent to pay.

## Payment evidence

| State             | Evidence                                                  | Permitted claim                        |
|-------------------|-----------------------------------------------------------|----------------------------------------|
| Amount selected   | Local selection event                                     | The user selected an amount            |
| Checkout created  | Provider checkout identifier                              | Checkout was opened or prepared        |
| Payment pending   | Provider reports a nonfinal state                         | Payment is being confirmed             |
| Payment confirmed | Signed provider result or verified server query           | The contribution completed             |
| Payment failed    | Provider final failure                                    | Payment did not complete               |
| Payment cancelled | Provider cancellation or user return without confirmation | No contribution recorded               |
| Payment refunded  | Provider refund record                                    | The contribution is no longer retained |

Client navigation, a success looking URL, or an open browser window is not confirmation. Provider secrets remain outside distributed applications.

## Conversion funnel

| Event             | Definition                                                               |
|-------------------|--------------------------------------------------------------------------|
| eligible          | The local policy allows an automatic presentation at this moment         |
| presented         | The support page became visible and usable                               |
| option selected   | The user chose an amount                                                 |
| wallet intent     | The user explicitly continued from amount review toward a payment method |
| checkout created  | The payment provider created a checkout                                  |
| checkout opened   | The user reached the provider                                            |
| payment confirmed | The provider verified completion                                         |
| dismissed         | The user left without selecting an amount                                |
| automatic opt out | The user disabled future automatic presentations                         |

## Metric definitions

| Metric                       | Numerator                                         | Denominator     | Required dimensions               |
|------------------------------|---------------------------------------------------|-----------------|-----------------------------------|
| Presentation rate            | Presented                                         | Eligible        | Product, version, edition, period |
| Selection rate               | Option selected                                   | Presented       | Product, page variant, period     |
| Checkout completion          | Payment confirmed                                 | Checkout opened | Provider, currency, period        |
| Contribution conversion      | Payment confirmed                                 | Presented       | Product, variant, period          |
| Gross value per presentation | Confirmed gross amount                            | Presented       | Currency, product, period         |
| Net value per presentation   | Confirmed net amount after known fees and refunds | Presented       | Currency, product, period         |
| Permanent opt out rate       | Automatic opt out                                 | Presented       | Product, variant, period          |

Rates are not reported without their counts, observation period, product, edition, and variant. Monetary values in different currencies remain separate unless an approved exchange rate and conversion time are recorded.

## Privacy preserving measurement

Local usage facts do not need to leave the device. If aggregate funnel measurement is approved, the reporting event contains the funnel step, product, release, edition, page variant, broad currency or market, and a random experiment assignment. It does not contain the facts shown on the page.

| Permitted when approved                               | Excluded                                             |
|-------------------------------------------------------|------------------------------------------------------|
| Product and release                                   | Save files or campaign history                       |
| Edition and distribution channel                      | Game commands or communications                      |
| Page and experiment variant                           | Books, documents, student work, grades, or filenames |
| Funnel event                                          | Exact location                                       |
| Contribution tier and currency                        | Hardware fingerprint                                 |
| Confirmed gross and net value from the payment system | Local session timestamps or detailed usage history   |

## Experiment discipline

Every conversion change starts as a written hypothesis. The experiment changes one principal factor, defines the eligible population, records assignment before presentation, and names the success and trust measures before results are examined.

| Field         | Meaning                                                        | Source of truth     |
|---------------|----------------------------------------------------------------|---------------------|
| experimentId  | Stable identifier                                              | Experiment registry |
| hypothesis    | Expected change and reason                                     | Business decision   |
| population    | Eligible products, editions, markets, and versions             | Experiment design   |
| variants      | Control and treatment with exact differences                   | Experiment design   |
| primaryMetric | One declared decision metric                                   | Metric registry     |
| guardrails    | Opt out, dismissal, complaint, refund, or other trust measures | Metric registry     |
| startAndStop  | Dates or evidence based stopping rule                          | Experiment design   |
| result        | Counts, rates, uncertainty, exclusions, and decision           | Analysis record     |

The system does not ship statements such as improves conversion, users prefer, or most popular until a completed analysis supports them for the named product and population.

## Initial hypotheses

| Hypothesis                                                      | Test                                                                                               | Primary measure                   | Guardrails                                              |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------|
| Verified usage facts increase relevance                         | Facts page versus generic support explanation                                                      | Confirmed value per presentation  | Dismissal and opt out                                   |
| A clear full meal recommendation reduces indecision             | Recommended meal versus equal visual weighting                                                     | Confirmed value per presentation  | Checkout completion and dismissal                       |
| A dated United States price study increases confidence          | Audited dated reference versus a contribution amount without the food evidence                     | Contribution conversion           | Disclosure engagement, complaints, and amount confusion |
| The fry calculation increases engagement without reducing value | Expandable calculation versus no fry option                                                        | Confirmed value per presentation  | Low value checkout failure and time to checkout         |
| Wallet commitment supports a practical meal sized ask           | After explicit wallet intent, meal recommendation versus retaining the smallest selected reference | Confirmed value per wallet intent | Cancellation, complaints, and amount confusion          |
| A completed activity is a better presentation moment            | Post completion versus next voluntary visit to Support                                             | Contribution conversion           | Opt out and complaint rate                              |

These are hypotheses. The document assigns no expected lift and contains no fabricated benchmark.

## Operating checks

| Frequency           | Check                                                                                                          | Evidence retained                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Each release        | Session definitions, build edition, payment destination, privacy copy, keyboard flow                           | Release checklist and test results                  |
| Price study         | Population, sample frame, restaurants, channel, product mapping, price, serving size, source, date, exclusions | Versioned observations and reproducible calculation |
| Fry estimate update | Scale method, sample count, combined mass, fragments policy, market, date                                      | Measurement record                                  |
| Provider change     | Minimum, currencies, fees, confirmation, refund, cancellation, data handling                                   | Approved provider review                            |
| Experiment close    | Population, counts, exclusions, result, guardrails, decision                                                   | Immutable experiment report                         |
| Periodic            | Stale prices, failed checkouts, opt outs, refunds, accessibility regressions                                   | Operations report                                   |

## Implementation acceptance tests

- A missing fact cannot render as zero.
- A partial history displays its coverage boundary.
- Background, paused, disconnected, or idle time is excluded according to the product definition.
- A crash adds no time after the last durable checkpoint.
- An outdated price study is never labeled current.
- A fry calculation rejects mismatched serving sizes and markets.
- The expanded fry calculation reproduces the stored result from its displayed inputs.
- A value below the provider minimum cannot create an invalid checkout.
- Opening checkout does not record a completed payment.
- Paid editions suppress automatic requests by default.
- Permanent opt out survives restart and upgrade.
- The page remains usable when pricing, network access, or the payment provider is unavailable.
- No application ships payment secrets or uploads product facts through the support module.

## Required decisions

| Decision                 | Evidence required before approval                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Payment provider         | Controlled account, supported markets and currencies, fees, minimums, confirmation method, refunds, data terms                                            |
| Price study              | Authorized sources, United States sampling frame, observation date, item identity, restaurant and channel specificity, calculation, and retained evidence |
| Fry serving              | Market specific item name, size, serving mass, price availability                                                                                         |
| Fry mass estimate        | Documented measurement method and retained samples or observations                                                                                        |
| Eligibility and cooldown | Observed product use and controlled experiment results                                                                                                    |
| Recommended contribution | Dated United States price rule and completed payment results                                                                                              |
| Analytics                | Explicit data inventory, consent basis, retention, deletion, and security review                                                                          |

## Legal and claims review

Before publication, qualified review must cover claims, the price study, checkout, privacy, tax, trademark, and storefront rules. Every objective claim needs retained evidence; correct arithmetic alone is not legal approval.

- Describe a sampled United States price as a study result. State its date, sample, statistic, product, channel, tax treatment, exclusions, units, and arithmetic; retain the evidence.
- Keep the actual charge distinct from the food comparison. Do not imply affiliation, endorsement, charitable status, or tax deductibility without authority; re-review material changes.